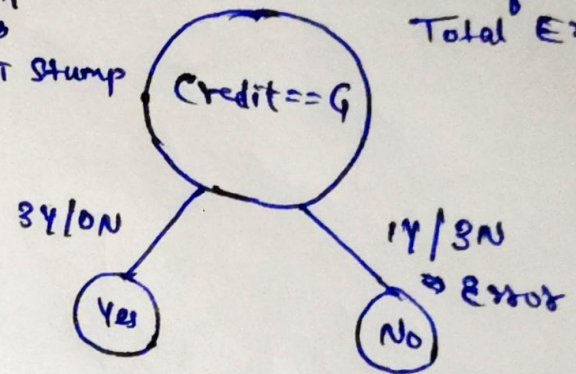


②. Sum of the Total Error and Performance of Stump.

Salary	Credit	Approval	Sample weight
$\leq 50K$	B	No	$1/7$
$\leq 50K$	G	Yes	$1/7$
$\leq 50K$	G	Yes	$1/7$
$> 50K$	B	No	$1/7$
$> 50K$	G	Yes	$1/7$
$> 50K$	N	Yes	$1/7$
$\leq 50K$	N	No	$1/7$

M_1
or DT Stump



Sum of all the Total Error = $1/7$

$$\therefore \ln[6] = \log_e(6) \approx 1.791759$$

$$\begin{aligned} \text{Performance of Stump} &= \frac{1}{2} \ln \left[\frac{1 - TE}{TE} \right] \\ &= \frac{1}{2} \ln \left[\frac{1 - 1/7}{1/7} \right] = \frac{1}{2} \ln[6] = \frac{1}{2} \times 1.791759 \\ &= 0.8958795 \\ &\approx \underline{0.896} \end{aligned}$$

Performance of Stump
 ≈ 0.896

$$f = \alpha_1(M_1) + \alpha_2(M_2) + \dots + \alpha_n(M_n)$$

$$\boxed{\alpha_1 = 0.896} \Rightarrow \text{weight}$$

